Lab 1

Project Scenario

The project aims to analyze the relationship between patients' mental health status and the use of associated medications, as well as to examine the role of health insurance systems in enhancing access to therapeutic services and psychological support.

لجين القرني 446013061

لمار مايت 44610449

ريماس الربيعي 446006803

Template 1: Data Source Description Table

Source Name:	Type:	Format:	Records:	Notes:
Mental Health Dataset	Structured	CSV	10,000	Contains mental health data such as stress, depression, anxiety, sleep hours, and productivity
Medical Insurance Dataset	Structured	CSV	1338	Contains individual-level medical insurance data including personal attributes and insurance charges
Mental Health Diagnosis and Treatment Monitoring Dataset	Structured	CSV	500	Contains patient-level mental health diagnosis, treatment details, symptom severity, and treatment outcomes.

Template 2: Data Profiling Table

The First Data Source (Mental Health Dataset)

Attribute	Data Type	Null %	Duplicates	Comments
Patient_ID	Integer	0%	No	Unique identifier for each individual (Primary Key)
age	Integer	0%	Yes	Age of the person
gender	String	0%	Yes	Male or Female
stress_level	Integer	0%	Yes	Stress level from 1 to 10
depression_score	Integer	0%	Yes	Depression level
anxiety_score	Integer	0%	Yes	Anxiety level
sleep_hours	Float	0%	Yes	Average sleep hours

The Second Data Source (Medical Insurance Dataset)

Attribute	Data type	Null%	Duplicates	Comments
Insurance_ID	Integer	0%	No	Primary Key – unique identifier for each record
age	Integer	0%	Yes	Age values may repeat across different individuals
sex	Categorical	0%	Yes	Gender values are repeated as expected
bmi	Float	0%	Yes	Body Mass Index values may be similar for multiple individuals
children	Integer	0%	Yes	Number of children can repeat for different records
smoker	Categorical	0%	Yes	Smoking status is repeated across individu
region	Categorical	0%	Yes	Regions are shared by multiple individu
charges	Float	0%	No	Insurance charges are mostly unique but may occasionally repeat

The Third Data Source (Mental Health Diagnosis and Treatment Monitoring Dataset)

Attribute	Data Type	Null %	Duplicates	Comments
Patient ID	Integer	0%	No	Patient ID numbers
age	Integer	0%	Yes	Indicates patient age
gender	String	0%	Yes	Indicates patient gender
Diagnosis	String	0%	Yes	Indicates mental health diagnosis
Symptom Severity	Integer	0%	Yes	Indicates symptom severity level
Mood Score	Integer	0%	Yes	Indicates mood level
Therapy Type	String	0%	Yes	Indicates therapy approach used

Lab 2: Structured vs. Unstructured Data

لمار محمد مايت 44610449
لجين القرني 446013061
ريماس الربيعي 446006803

1) Unstructured Data Analysis Table

Data Source Name: Patient Insurance Narratives

Type: Unstructured Text

Attribute	Data Type	Structured / Unstructured	Justification
Narrative_Text	Text	Unstructured	Free text written by patients
Disease	Text	Unstructured	Mentioned inside paragraph
Employment_Status	Text	Unstructured	Appears in text
Treatment	Text	Unstructured	Therapy or medication mentioned
Cost_Before_Insurance	Text	Unstructured	Written inside sentence
Insurance_Type	Text	Unstructured	Appears inside text
Cost_After_Insurance	Text	Unstructured	Extracted from text
Improvement	Text	Unstructured	Patient condition after treatment

2) Transformation Plan Table

Step	Technique	Expected Structured Output
1	Remove emojis & clean text	Clean narrative text
2	Identify keywords	Disease name
3	Extract numbers	Cost before & after insurance

4	Identify employment status	Student / Employee
5	Classify insurance type	Government / Private / Student
6	Identify treatment type	Therapy / Medication
7	Store data in table	Structured dataset

3) Proposed Structured Output Table

Table Name: Patient_Insurance_Data

Field Name	Data Type	Source	Description
Record_ID	Integer	Generated	Unique ID
Disease	Varchar	From text	Type of illness
Employment_Status	Varchar	From text	Student or Employee
Treatment_Type	Varchar	From text	Therapy or Medication
Cost_Before	Decimal	Extracted	Cost before insurance
Insurance_Type	Varchar	From text	Type of insurance
Cost_After	Decimal	Extracted	Cost after insurance
Improvement_Status	Varchar	From text	Patient condition after treatment

Lab 3: Data preprocessing & Cleaning for Integration

Data Quality Assessment Table

44610449 لمار محمد مايت

446013061لجين القرني

ريماس عبدالهادي الربيعي 446006803

Dataset 1: Medical Insurance

Attribute	Problem Type	Issue Description	Impact	Source
age	Integrity Violation	Presence of unrealistic values (e.g., negative or extremely high ages)	Leads to incorrect analysis and unreliable predictions	Medical Insurance Dataset
gender	Semantic Issue	Inconsistent category values (e.g., male, Male, M)	Causes incorrect grouping and duplication of categories	Medical Insurance Dataset
charges	Integrity Violation	Presence of extreme outliers in cost values	Skews statistical analysis and model performance	Medical Insurance Dataset

Dataset 2: Mental Health (Survey Dataset)

Attribute	Problem Type	Issue Description	Impact	Source
timestamp	Format Inconsistency	Different date/time formats across records	Complicates time-based analysis	Mental Health Dataset
gender	Semantic Issue	Inconsistent and diverse entries (Male, male, Female, Other)	Reduces data quality and affects classification	Mental Health Dataset
country	Structural Issue	Country names written in	Causes issues in geographic analysis	Mental Health Dataset

		different formats or spellings		

Dataset 3: Mental Health Diagnosis & Treatment Monitoring

Attribute	Problem Type	Issue Description	Impact	Source
patient_id	Structural Issue	Possible duplicate IDs	Affects record uniqueness and data integrity	Mental Health Monitoring Dataset
treatment_start_date	Format Inconsistency	Inconsistent date formats	Difficult to calculate treatment duration	Mental Health Monitoring Dataset
symptom_severity	Integrity Violation	Values outside expected range (e.g., not between 1–10)	Produces inaccurate severity analysis	Mental Health Monitoring Dataset

Dataset 1: Medical Insurance - Cleaning Strategy Table

Attribute	Problem	Cleaning Method	Rule
age	Unrealistic values	Remove or replace with median	0–100
gender	Inconsistent values	Standardize to Male/Female	Consistent format
charges	Outliers	Use IQR to cap/remove	Within range

Dataset 2: Mental Health Survey - Cleaning Strategy Table

Attribute	Problem	Cleaning Method	Rule
timestamp	Different formats	Convert to YYYY-MM-DD	Unified format
gender	Inconsistent entries	Normalize values	Predefined categories
country	Different spellings	Standardize names	Consistent naming

Dataset 3: Mental Health Diagnosis & Treatment - Cleaning Strategy Table

Attribute	Problem	Cleaning Method	Rule
patient_id	Duplicate IDs	Remove duplicates	Unique ID
treatment_start_date	Different formats	Convert to YYYY-MM-DD	Unified format
symptom_severity	Out of range	Fix/remove invalid values	1–10

Dataset 1 Medical Insurance

Attribute	Null %	Duplicates	Pattern Issues	Key Candidate
Age	**0**	**129**	**None**	**No**
Sex	**0**	**1336**	**None**	**No**
Bmi	**0**	**1063**	**None**	**No**
Children	**0**	**1332**	**None**	**No**
Smoker	**0**	**1336**	**None**	**No**
Region	**0**	**1334**	**None**	**No**
Expenses	**0**	**1**	**None**	**No**

Dataset 2 Mental Health

Attribute	Null %	Duplicates	Pattern Issues	Key Candidate
Age	**0**	**9952**	**None**	**No**
Gender	**0**	**9996**	**None**	**No**
employment_status	**0**	**9996**	**None**	**No**
work_environment	**0**	**9997**	**None**	**No**
mental_health_history	**0**	**9998**	**None**	**No**
seeks_treatment	**0**	**9990**	**None**	**No**
stress_level	**0**	**9929**	**None**	**No**
sleep_hours	**0**	**9992**	**None**	**No**
physical_activity_days	**0**	**9969**	**None**	**No**
depression_score	**0**	**9978**	**None**	**No**
anxiety_score	**0**	**9978**	**None**	**No**
social_support_score	**0**	**9899**	**None**	**No**
productivity_score	**0**	**9454**	**None**	**No**
mental_health_risk	**0**	**9997**	**None**	**No**

Dataset 3 Mental Health Diagnosis & Treatment Monitoring

Attribute	Null %	Duplicates	Pattern Issues	Key Candidate
Patient_ID	**0**	**0**	**None**	**Yes**
Age	**0**	**457**	**None**	**No**
Gender	**0**	**498**	**None**	**No**
Diagnosis	**0**	**496**	**None**	**No**

Symptom Severity (1-10)	**0**	**494**	**None**	**No**
Mood Score (1-10)	**0**	**494**	**None**	**No**
Sleep Quality (1-10)	**0**	**494**	**None**	**No**
Physical Activity (hrs/week)	**0**	**490**	**None**	**No**
Medication	**0**	**494**	**None**	**No**
Therapy Type	**0**	**496**	**None**	**No**
Treatment Start Date	**0**	**399**	**None**	**No**
Treatment Duration (weeks)	**0**	**491**	**None**	**No**
Stress Level (1-10)		**494**	**None**	**No**
Outcome	**0**	**497**	**None**	**No**
reatment Progress (1-10)	**0**	**494**	**None**	**No**
AI-Detected Emotional State	**0**	**494**	**None**	**No**
Adherence to Treatment (%),	**0**	**469**	**None**	**No**

Lab 4

Selected Data Sources

- Source A (df1): Mental Health Diagnosis and Treatment Monitoring
- Source B (df3): Medical Insurance Dataset

Lujain Alqarni 446013061

Lamar mayet 44610449

ريماس عبدالهادي الربيعي 446006803

Source A Attribute	Source B Attribute	Match Type	Confidence	Justification
Age	Age	Match	High	Exact same attribute representing age
Gender	Sex	Match	High	Different names but identical meaning
Patient ID	-	No match	Low	No corresponding identifier in Source B
Diagnosis	-	No match	Low	No equivalent medical diagnosis field
Symptom Severity	-	No match	Low	No comparable attribute in Source B
Mood Score	-	No match	Low	No related variable in Source B
Sleep Quality	-	No match	Low	No equivalent attribute
Physical Activity	-	No match	Low	No direct corresponding column
Medication	-	No match	Low	No equivalent attribute in insurance dataset
Therapy Type	-	No match	Low	No corresponding column
Treatment Start Date	-	No match	Low	No date-related field in Source B
Treatment Duration	-	No match	Low	Not directly comparable to any attribute
Stress Level	-	No match	Low	No corresponding variable

Outcome	-	No match	Low	No equivalent attribute
Treatment Progress	-	No match	Low	No corresponding field
AI Emotional State	-	No match	Low	No equivalent in Source B
Adherence	-	No match	Low	No corresponding attribute

Summary

- Match: 2 attributes
- Partial Match: 0 attributes
- No Match: 15 attributes

- Lab: Lab 5 – Schema Mappings

Lujain Alqarni ID: 44601306

Lamar Mayet 44610449

Selected Data Sources

- Source A (df1): Mental Health Diagnosis and Treatment Monitoring
- Source B (df3): Medical Insurance Dataset

Attributes of Source A (df1)

- Patient ID
- Age
- Gender
- Diagnosis
- Symptom Severity
- Mood Score
- Sleep Quality
- Physical Activity
- Medication
- Therapy Type
- Treatment Start Date
- Treatment Duration
- Stress Level
- Outcome
- Treatment Progress
- AI Emotional State
- Adherence

Source B (df3) Attributes

- Age
- Sex
- BMI
- Children
- Smoker
- Region
- Charges

Table 1: Schema Mapping Table

Source Attribute	Target Attribute	Mapping Type	Short Description
Age	Age	1:1	Direct mapping of the same attribute representing age in both datasets
Gender	Sex	1:1	Some concept with different attribute names (gender representation)
Patient ID	-	Conditional	No direct mapping; may be used as an internal identifier only in Source A

Table 2: Three mapping rules.

Rule	Rule Expression	Explanation
1	If Age exists → map directly	Same attribute in both datasets
2	If Gender = Male/Female → map to Sex	Same meaning with different naming
3	If Diagnosis varies → Charges may vary	Different diagnoses affect cost

Lab6 : Data Warehouse design and Data Marts

Lujain Alqarni (44601306)
Lamar Mayet (44610449)
Remas Alroubie (446006803)

Template 1 – Star Schema Design Table

Table type	Table name	Key(s)	Main attribute(Summary)	Description
Fact	FACT_Mental_Health	Patient_ID (FK), Treatment_ID (FK)	Visit_Count, Cost, Outcome_Score, Insurance_Coverage	Stores quantitative information about each patient's mental health visits, treatments, and associated costs

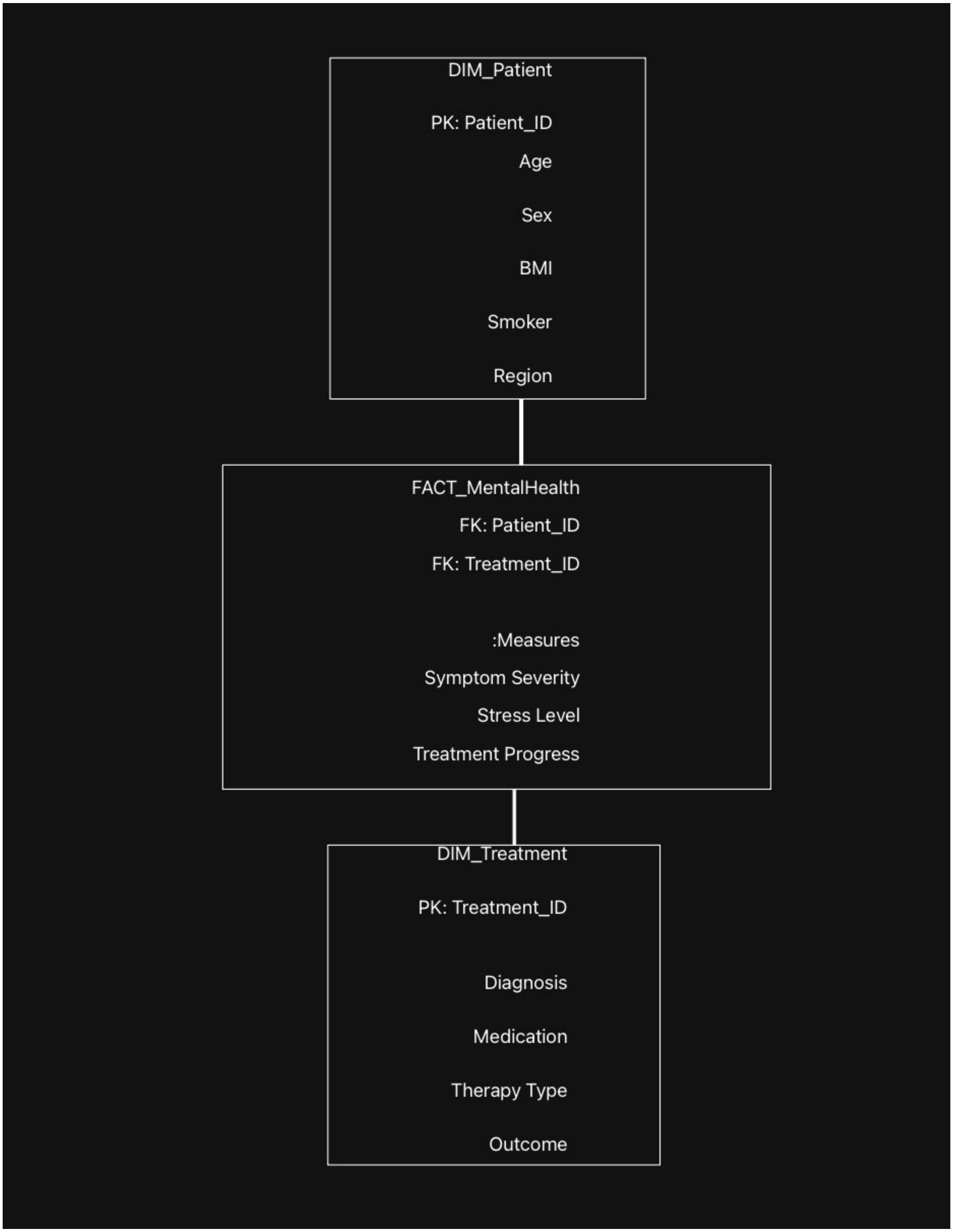
Dimension Table

Table type	Table name	Key(s)	Main attribute	Description
Dimension	DIM_Patient	Patient_ID (PK)	Age, Sex, BMI, Smoker, Region	Stores patient demographic information
Dimension	DIM_Treatment	Treatment_ID (PK)	Diagnosis, Medication, Therapy Type, Outcome	Stores treatment and diagnosis details

Template 2 – Data Loading Log

Table name	Row inserted(Number)	Source used	Notes / Simplefictions
DIM_Patient	10	Medical Insurance Dataset	Age, Sex, BMI, Smoker, Region
DIM_Treatment	10	Mental Health Diagnosis Dataset	Diagnosis, Medication, Therapy, Outcome
FACT_MentalHealth	10	All The 3 Source	Severity, Stress, Progress کـ measures

Star Schema Design Diagra



```sql
-- إنشاء قاعدة بيانات جديدة
CREATE DATABASE Lab6DB;
USE Lab6DB;

-- Dimension Tables
CREATE TABLE DIM_Patient (
    Patient_ID INT PRIMARY KEY,
    Age INT,
    Sex VARCHAR(10),
    BMI FLOAT,
    Smoker BOOLEAN,
    Region VARCHAR(50)
);

CREATE TABLE DIM_Treatment (
    Treatment_ID INT PRIMARY KEY,
    Diagnosis VARCHAR(100),
    Medication VARCHAR(100),
    Therapy_Type VARCHAR(50),
    Outcome VARCHAR(50)
);
```

```sql

-- Fact Table
CREATE TABLE FACT_MentalHealth (
    Fact_ID INT PRIMARY KEY,
    Patient_ID INT,
    Treatment_ID INT,
    Visit_Count INT,
    Cost DECIMAL(10,2),
    Outcome_Score FLOAT,
    Insurance_Coverage VARCHAR(50),
    FOREIGN KEY (Patient_ID) REFERENCES DIM_Patient(Patient_ID),
    FOREIGN KEY (Treatment_ID) REFERENCES DIM_Treatment(Treatment_ID)
);

```

✔	11	21:41:59	CREATE DATABASE Lab6DB	1 row(s) affected	0.046 sec
✔	12	21:41:59	USE Lab6DB	0 row(s) affected	0.000 sec
✔	13	21:41:59	CREATE TABLE DIM_Patient (Patient_ID INT PRIMARY KEY, Age INT, Se...	0 row(s) affected	0.125 sec
✔	14	21:41:59	CREATE TABLE DIM_Treatment (Treatment_ID INT PRIMARY KEY, Diagnosi...	0 row(s) affected	0.078 sec
✔	15	21:42:00	CREATE TABLE FACT_MentalHealth (Fact_ID INT PRIMARY KEY, Patient_I...	0 row(s) affected	0.125 sec

Lab 7 : Data Warehouse design and Data Marts

Lujain Alqarni (44601306)

Lamar Mayet (44610449)

Remas Alroubie (446006803)

Template A – Query Log:

Query	Business Question	Tool	Result Summary
1	Which diagnosis has the highest average stress level?	SQL	Major Depressive Disorder had the highest avg stress (6.83) with 6 patients
2	Which therapy type has the best treatment progress?	SQL	Mindfulness-Based Therapy showed highest avg progress (7.50)
3	Which region has the highest average BMI?	SQL	Southeast region had the highest avg BMI (31.47) with 4 patients
4	What is the relationship between depression score and treatment progress?	SQL	Low Depression group had the best avg treatment progress (7.50)

```sql
USE Lab6DB;

-- ============================================================
-- Query 1:
-- Which diagnosis is associated with the highest average stress level?
-- أي تشخيص يرتبط بأعلى مستوى ضغط نفسي؟
-- ============================================================
SELECT
    t.diagnosis,
    AVG(f.stress_level)        AS avg_stress,
    AVG(f.symptom_severity)    AS avg_severity,
    COUNT(*)                   AS total_patients
FROM FACT_MentalHealth f
JOIN DIM_Treatment t ON f.treatment_id = t.treatment_id
GROUP BY t.diagnosis
ORDER BY avg_stress DESC;

-- ============================================================
-- Query 2:
-- Which therapy type has the best treatment outcome (Improved)?
-- أي نوع علاج يعطي أفضل نتيجة؟
-- ============================================================
SELECT
    t.therapy_type,
    t.outcome,
    COUNT(*)                                      AS total_cases,
    AVG(f.treatment_progress)                     AS avg_progress,
    SUM(CASE WHEN t.outcome = 'Improved' THEN 1 ELSE 0 END) AS improved_count
FROM FACT_MentalHealth f
JOIN DIM_Treatment t ON f.treatment_id = t.treatment_id
GROUP BY t.therapy_type, t.outcome
ORDER BY avg_progress DESC;

-- ============================================================
-- Query 3:
-- Which region has the highest average BMI among patients?
-- أي منطقة فيها أعلى متوسط BMI؟
-- ============================================================
SELECT
    p.region,
    AVG(p.bmi)              AS avg_bmi,
    COUNT(*)                AS total_patients,
    SUM(CASE WHEN p.smoker = 'yes' THEN 1 ELSE 0 END) AS smokers_count
FROM FACT_MentalHealth f
JOIN DIM_Patient p ON f.patient_id = p.patient_id
GROUP BY p.region
ORDER BY avg_bmi DESC;

-- ============================================================
-- Query 4:
-- What is the relationship between depression score and treatment progress?
-- ما العلاقة بين درجة الاكتئاب وتقدم العلاج؟
-- ============================================================
SELECT
    CASE
        WHEN f.depression_score >= 25 THEN 'High Depression (25+)'
        WHEN f.depression_score >= 15 THEN 'Medium Depression (15-24)'
        ELSE 'Low Depression (0-14)'
    END AS depression_level,
    COUNT(*)                   AS total_patients,
    AVG(f.treatment_progress)  AS avg_treatment_progress,
    AVG(f.stress_level)        AS avg_stress,
    AVG(f.anxiety_score)       AS avg_anxiety
FROM FACT_MentalHealth f
GROUP BY depression_level
ORDER BY avg_treatment_progress DESC;
```

Data Analysis and Insights

Analysis – Query 1

Theresultsindicatethat Major Depressive Disorder is associated with the highest average stress level among patients, with an average stress score of approximately 6.83 across six patients.

Although Generalized Anxiety shows the highest average symptom severity, this result is based on a very small sample size, which limits its reliability.

Overall, the findings suggest that patients diagnosed with Major Depressive Disorder experience consistently higher stress levels, highlighting the need for targeted stress management strategies for this group.

Analysis – Query 2

The analysis shows that Mindfulness-Based Therapy has the highest average treatment progress score)7.5(among all therapy types, indicating better improvement compared to other treatments.

In contrast ,Dialectical Behavioral Therapy demonstrates the lowest average treatment progress, suggesting limited effectiveness within this dataset.

It is also observed that none of the therapy types recorded cases classified as Improved, which may reflect challenges in achieving positive treatment outcomes or limitations due to the small sample size.

Analysis – Query 3

The results reveal that the Southeast region has the highest average BMI among patients, with an average value of approximately ,31.48 indicating a higher prevalence of elevated body mass index in this region.

On the other hand, the Northwest region shows the lowest average BMI, suggesting relatively healthier BMI levels among patients in that area.

Smoking behavior appears limited in this dataset, with only one smoker identified in the Southwest region. However, the small sample size should be considered when interpreting these findings.

Analysis – Query 4

Theanalysis demonstrates a clear relationship between depression severity and treatment progress .Patients in the Low Depression category show the highest average treatment progress, indicating better recovery outcomes .

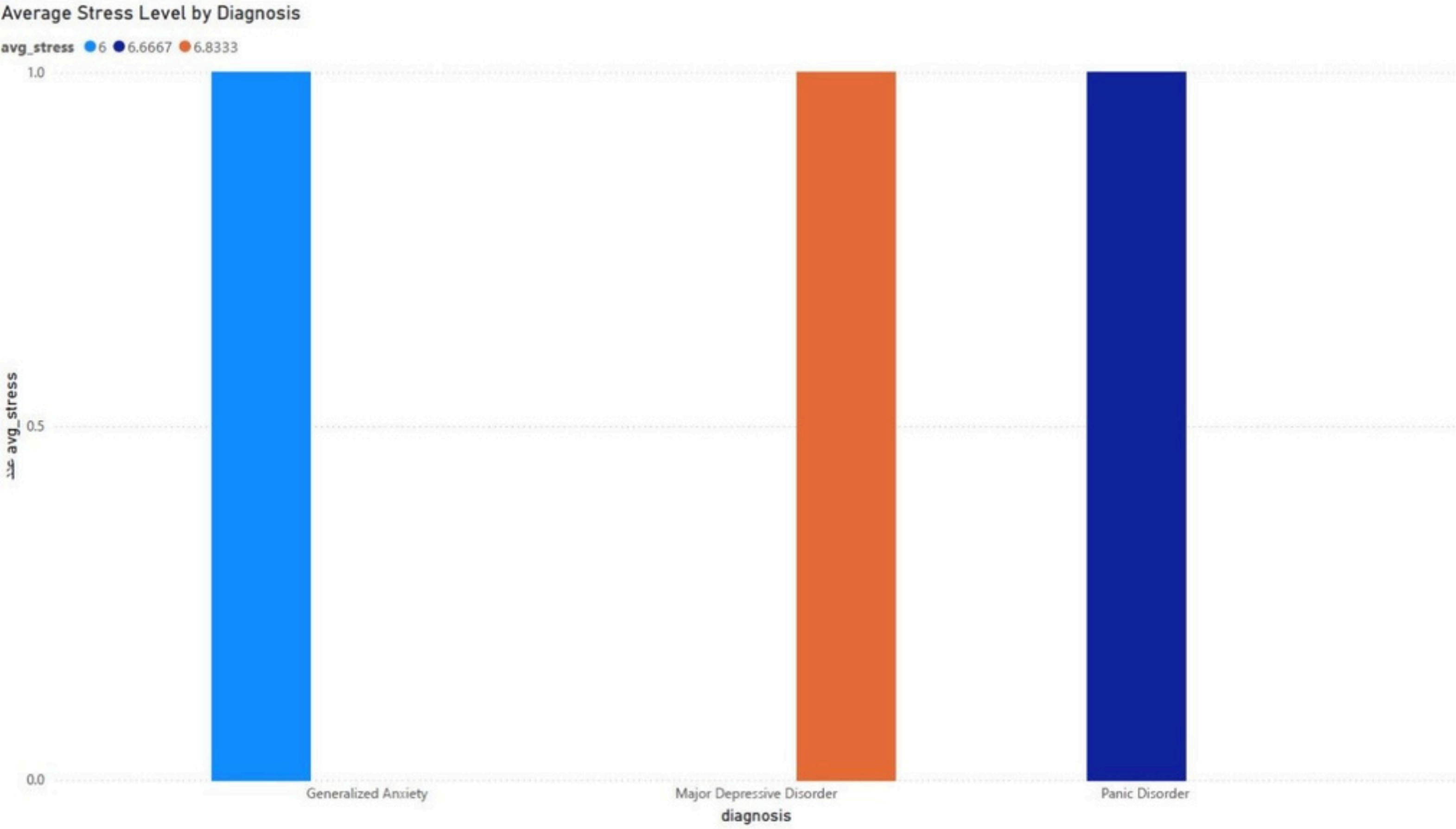
In contrast, patients with Medium Depression exhibit the lowest treatment progress alongside the highest anxiety levels, suggesting that moderate depression combined with high anxiety may negatively impact treatment effectiveness.

Although patients with High Depression show slightly better progress than the medium group, their improvement remains lower than those with low depression, emphasizing the importance of early intervention.

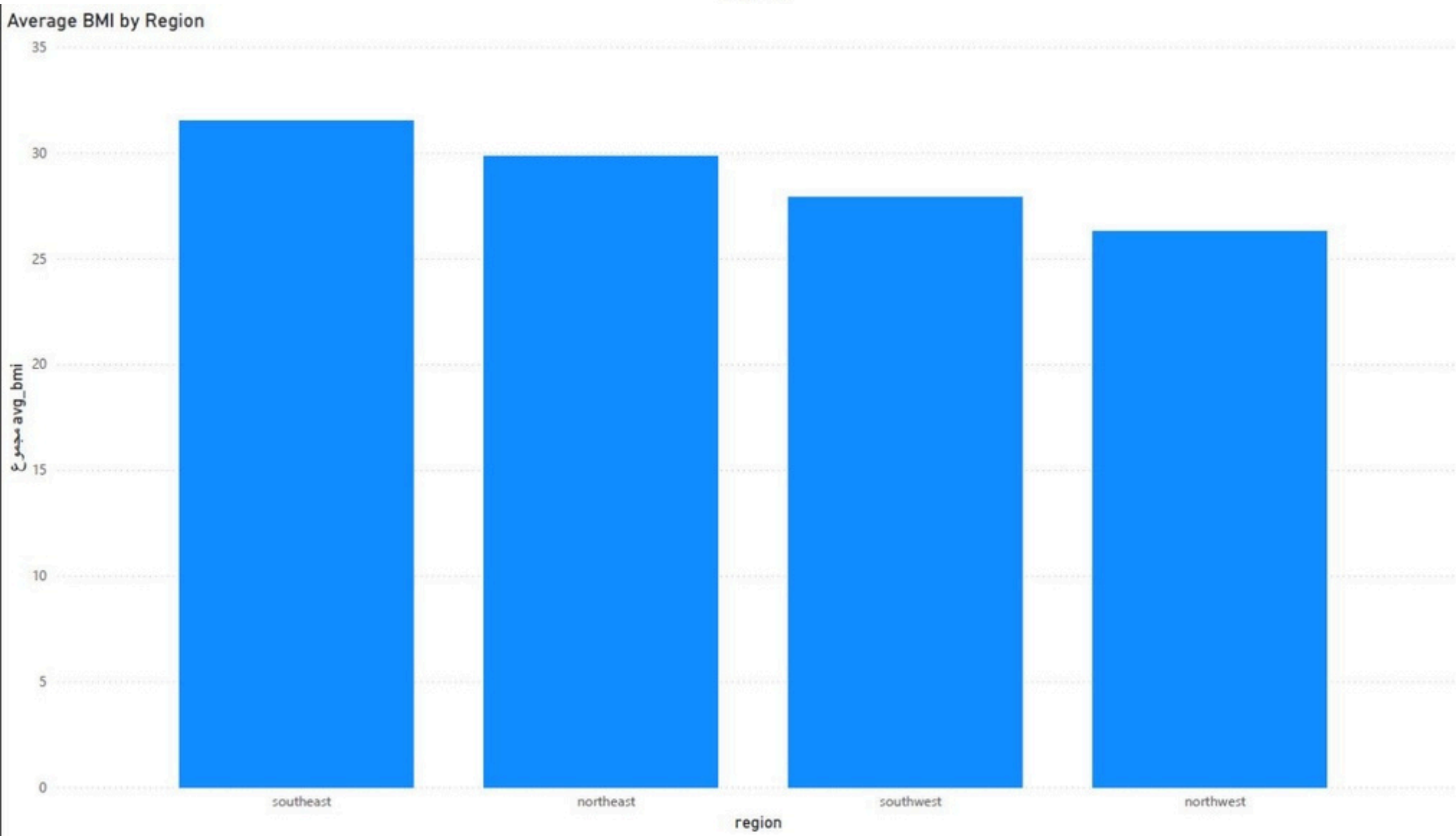
Overall Summary

Overall, the analysis highlights important patterns across diagnosis, treatment types, regions, and depression severity levels. The findings emphasize that stress levels, treatment effectiveness, and patient characteristics vary significantly and should be considered when designing personalized treatment plans.

These insights provide valuable support for decision-making in improving patient outcomes and optimizing mental health treatment strategies.

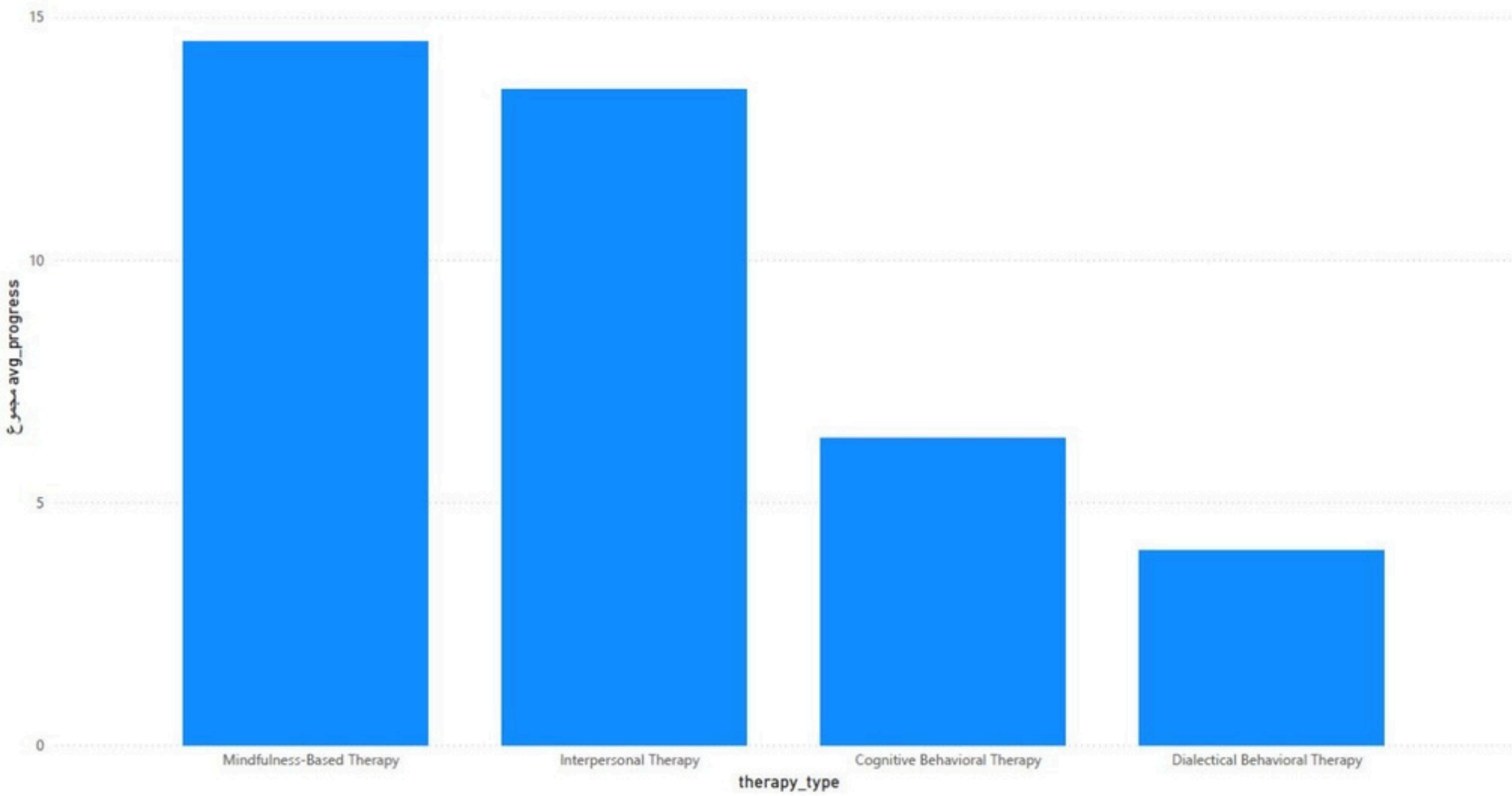


Average Stress Level by Diagnosis
avg_stress 6 6.6667 6.8333
1.0
0.5
0.0
عدد avg_stress
Generalized Anxiety
Major Depressive Disorder
Panic Disorder
diagnosis



Average BMI by Region
35
30
25
20
15
10
5
0
مجموع avg_bmi
southeast
northeast
southwest
northwest
region

Treatment Progress by Therapy Type



Depression Severity vs Treatment Progress